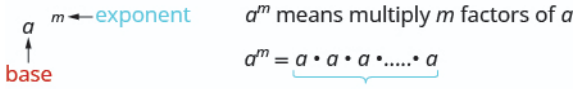
Chapter Review

Key Terms

binomial A binomial is a polynomial with exactly two terms.
degree of a constant The degree of a constant is 0.
degree of a polynomial The degree of a polynomial is the highest degree of all its terms.
degree of a term The degree of a term of a polynomial is the exponent of its variable.
greatest common factor The greatest common factor (GCF) of two or more expressions is the largest expression that is a factor of all the expressions.
monomial A term of the form ax^m, where a is a constant and m is a whole number, is called a monomial.
negative exponent If n is a positive integer and $a \neq 0$, then $a^{-n} = \frac{1}{a^n}$.
polynomial A polynomial is a monomial, or two or more monomials, combined by addition or subtraction.
scientific notation A number expressed in scientific notation when it is of the form $a \times 10^n$, where $a \geq 1$ and $a < 10$, and n is an integer.
trinomial A trinomial is a polynomial with exactly three terms.
zero exponent If a is a non-zero number, then $a^0 = 1$. Any nonzero number raised to the zero power is 1.

Key Concepts

10.2 Use Multiplication Properties of Exponents



- **Exponential Notation**
  This is read a to the mth power.
- **Product Property of Exponents**
  - If a is a real number and m, n are counting numbers, then
    $$a^m \cdot a^n = a^{m+n}$$
  - To multiply with like bases, add the exponents.
- **Power Property for Exponents**
  - If a is a real number and m, n are counting numbers, then
    $$\left(a^m\right)^n = a^{m \cdot n}$$
- **Product to a Power Property for Exponents**
  - If a and b are real numbers and m is a whole number, then
    $$(ab)^m = a^m b^m$$

10.3 Multiply Polynomials

- **Use the FOIL method for multiplying two binomials.**

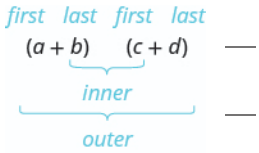
Step 1. Multiply the **First** terms.

Step 2. Multiply the **Outer** terms.

Step 3. Multiply the **Inner** terms.

Step 4. Multiply the **Last** terms.

Step 5. Combine like terms, when possible.



- **Multiplying Two Binomials:** To multiply binomials, use the:
  - Distributive Property
  - FOIL Method
  - Vertical Method
- **Multiplying a Trinomial by a Binomial:** To multiply a trinomial by a binomial, use the:
  - Distributive Property

- Vertical Method

10.4 Divide Monomials

- **Equivalent Fractions Property**
  - If $a,\ b,\ c$ are whole numbers where $b \neq 0,\ c \neq 0$, then
$$\frac{a}{b} = \frac{a \cdot c}{b \cdot c} \text{ and } \frac{a \cdot c}{b \cdot c} = \frac{a}{b}$$
- **Zero Exponent**
  - If a is a non-zero number, then $a^0 = 1$.
  - Any nonzero number raised to the zero power is 1.
- **Quotient Property for Exponents**
  - If a is a real number, $a \neq 0$, and $m,\ n$ are whole numbers, then
$$\frac{a^m}{a^n} = a^{m-n}, \quad m > n \quad \text{and} \quad \frac{a^m}{a^n} = \frac{1}{a^{n-m}}, \quad n > m$$
- **Quotient to a Power Property for Exponents**
  - If a and b are real numbers, $b \neq 0$, and m is a counting number, then
$$\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$$
  - To raise a fraction to a power, raise the numerator and denominator to that power.

10.5 Integer Exponents and Scientific Notation

- **Summary of Exponent Properties**
  - If a, b are real numbers and m, n are integers, then

Product Property	$a^m \cdot a^n = a^{m+n}$
Power Property	$(a^m)^n = a^{m \cdot n}$
Product to a Power Property	$(ab)^m = a^m b^m$
Quotient Property	$\frac{a^m}{a^n} = a^{m-n},\ a \neq 0$
Zero Exponent Property	$a^0 = 1,\ a \neq 0$
Quotient to a Power Property	$\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m},\ b \neq 0$
Definition of Negative Exponent	$a^{-n} = \frac{1}{a^n}$

- **Convert from Decimal Notation to Scientific Notation:** To convert a decimal to scientific notation:
  Step 1. Move the decimal point so that the first factor is greater than or equal to 1 but less than 10.
  Step 2. Count the number of decimal places, n, that the decimal point was moved.
  Step 3. Write the number as a product with a power of 10.
    - If the original number is greater than 1, the power of 10 will be 10^n.
    - If the original number is between 0 and 1, the power of 10 will be 10^{-n}.

  Step 4. Check.
- **Convert Scientific Notation to Decimal Form:** To convert scientific notation to decimal form:
  Step 1. Determine the exponent, n, on the factor 10.
  Step 2. Move the decimal n places, adding zeros if needed.
    - If the exponent is positive, move the decimal point n places to the right.
    - If the exponent is negative, move the decimal point $|n|$ places to the left.

  Step 3. Check.

10.6 Introduction to Factoring Polynomials

- **Find the greatest common factor.**
  Step 1. Factor each coefficient into primes. Write all variables with exponents in expanded form.
  Step 2. List all factors—matching common factors in a column. In each column, circle the common factors.
  Step 3. Bring down the common factors that all expressions share.
  Step 4. Multiply the factors.
- **Distributive Property**
  - If a, b, c are real numbers, then
    $a(b + c) = ab + ac$ and $ab + ac = a(b + c)$